# CURRICULUM VITAE SHIWEI LIU

## BACKGROUND


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| **Name** | Shiwei Liu |
| **Position** | Postdoctoral Fellow at University of Texas at Austin |
| **Web Page** | Google Scholar Personal Website |
| **Address** | 800 W 38th St, Austin, TX 78705, USA |
| **E-mail** | s.liu3@tue.nl, shiwei.liu@austin.utexas.edu |

## RESEARCH INTERESTS AND VISION

**The Science of Deep Neural Networks.** Despite the ever-strong results achieved by deep neural networks, how to fully understand the behavior of deep learning remains mysterious, resulting in potential risks to our society and industry. I study and understand the behavior of deep learning from a scientific perspective, by conducting empirical experiments in practice. I believe that we can enhance our understanding of neural networks by conducting scientific research on their practical behavior.

**Efficient Neural Network Training and Scaling.** Costs associated with training and deploying modern deep neural networks have increased exponentially, beyond the reach of hardware development. My research focuses on advancing sparsity as a promising solution to boost the trainability & scalability of neural networks.

## EDUCATIONAL HISTORY

**University of Texas at Austin (UT Austin), USA** *September 2022 - Present*
Postdoctoral Fellow (Machine Learning)
Department: Electrical and Computer Engineering & IFML
Supervisors: Dr. Atlas Wang

**Eindhoven University of Technology (TU/e), The Netherlands** *March 2018 - April 2022*
Doctor of Philosophy (Ph.D.) (Machine Learning)
Thesis: Sparse Neural Network Training with In-Time Over-Parameterization (**Cum Laude**)
Department: Mathematics and Computer Science
Promotors: Prof. Dr. Mykola Pechenizkiy; Dr. Decebal Constantin Mocanu

**Harbin Institute of Technology, China** *September 2015 - July 2017*
Master of Science (M.Sc.) (System Identification)
Department: Mechanical Engineering and Automation
Advisor: Prof. Dr. Ai-Guo Wu

**North University of China, China** *September 2011 - July 2015*
Bachelor of Engineering (B.Eng.) (Control Engineering)
Department: Mechanical Engineering and Automation
Advisor: Dr. Zhongqing Wang

## SELECTED RESEARCH & PUBLICATIONS

Shiwei Liu has co-authored over 20 papers in top-tier machine learning venues (NeurIPS, ICML, ICLR, IJCV, AAAI, IJCAI, ICCV, UAI, LoG, etc), with 19 first-author and 4 last-author papers.

[**1**] Tianjin Huang, Tianlong Chen, Meng Fang, Vlado Menkovski, Jiaxu Zhao, Lu Yin, Yulong Pei, Decebal Constantin Mocanu, Zhangyang Wang, Mykola Pechenizkiy, **Shiwei Liu**. *You Can Have*

*Better Graph Neural Networks by Not Training Weights at All: Finding Untrained Graph Tickets.* Learning on Graphs Conference (LoG), 2022. **Best Paper Award**. Paper Code

[2] **Shiwei Liu**, Lu Yin, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *Do We Actually Need Dense Over-Parameterization? In-Time Over-Parameterization in Sparse Training.* The Thirty-Eighth International Conference on Machine Learning (ICML), PMLR, 2021. Paper Code

[3] **Shiwei Liu***, Tianlong Chen*, Zhenyu Zhang, Xuxi Chen, Tianjin Huang, Ajay Kumar Jaiswal, and Zhangyang Wang. *Sparsity May Cry: Let Us Fail (Current) Sparse Neural Networks Together!* International Conference on Learning Representations (ICLR), 2023. **Top-25% Spotlight**. Paper Code

[4] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Xuxi Chen, Qiao Xiao, Boqian Wu, Mykola Pechenizkiy, Decebal Mocanu, and Zhangyang Wang. *More Convnets in the 2020s: Scaling Up Kernels Beyond 51x51 Using Sparsity.* International Conference on Learning Representations (ICLR), 2023. Paper Code

[5] Duc N.M Hoang, **Shiwei Liu**, Radu Marculescu, Zhangyang Wang. *Revisiting Pruning at Initialization Through the Lens of Ramanujan Graph.* International Conference on Learning Representations (ICLR), 2023. **Top-5% Spotlight**. Paper Code

## AWARDS AND HONOURS

- Best Paper Award, Learning on Graphs Conference (LoG 2022) *11/2022*
- AAAI-23 Travel Scholarship, USA *12/2022*
- IFML Postdoctoral Fellowship, USA *9/2022*
- Cum Laude (Distinguished Ph.D. thesis), Eindhoven University of Technology, Netherlands *4/2022*
- Outstanding Intern, JD Explore Academy, 2% in company *10/2021*
- The First Clase Scholarship, Harbin Institute of Technology, China *07/2017*
- China National Scholarship, China, 0.2% in China *10/2016*
- The First Clase Scholarship, Harbin Institute of Technology, China *07/2016*
- The Outstanding Graduate, North University of China, China *07/2015*

## GRANTS (TOTAL: $360,000)

- Shiwei Liu. (Lead Investigator), "Large-Scale Sparse Neural Models.", Sponsored by Dutch Research Council: NWO Grants for Computing Time on the National Computer Facilities, $220,000. Expected *April 2022 - April 2024*

- Shiwei Liu. (Single Lead Investigator), "Carnegie Bosch Fellowships", Sponsored by Carnegie Bosch Institute, *2023 - 2025 (declined)*

- Shiwei Liu. (Single Lead Investigator), "Postdoctoral Fellowship", Sponsored by Institute for Foundations of Machine Learning (IFML), $140,000. Expected *September 2022 - September 2024*

## INVITED TALKS

- CTSTA Workshop in PLDI 2023. Organizer: Fredrik Kjolstad (Stanford), Saman Amarasinghe (MIT), Michelle Mills Strout (University of Arizona) Jun. 2023
- EfficientML Reading Group. Organizer: Olga Saukh (TU Graz) May. 2023
- Sparsity Reading Group. Organizer: Anna Golubeva (MIT), Dan Alistarh (IST Austria), Decebal Constantin Mocanu (University of Luxembourg), Gintare Karolina Dziugaite (Google Brain, McGill University), Utku Evci (Google Brain), Yani Ioannou (University of Calgary) Mar. 2023
- Sharc Group, Organizer: Cong Hao (Georgia Institute of Technology) Mar. 2023
- IFML, Organizer: Adam Klivans (UT Austin), Alexandros Dimakis (UT Austin) Jan. 2023
- LoG2022 Meetup, Organizer: Jure Leskovec (Stanford), Kexin Huang (Stanford) Dec. 2022

- NCSU Reliable & Efficient Computing Lab, Organizer: Dongkuan Xu (NC State) Nov. 2022
- EIC Lab, Organizer: Yingyan Lin (Georgia Institute of Technology) Sep. 2022
- Techbeat Sep. 2022
- AI XinQingNian Nov. 2022
- Data Mining Group, Organizer: Mykola Pechenizkiy (TU Eindhoven) May. 2022

## WORK EXPERIENCE

**Postdoctoral Fellow at The University of Texas at Austin, USA** *September 2022 - Present*
- Study and understand the behavior of deep learning from a scientific perspective by conducting empirical experiments in practice.
- Scale sparsity up to large language models.
- Provide mentorship to PhD students.

**Ph.D. at Eindhoven University of Technology, Netherlands** *March 2018 - April 2022*
- Develop training efficient and scalable machine learning algorithms and models.
- Advance sparsity as a promising solution to improve the reliability of ML.

**Research Intern at JD Explore Academy** *June 2021 - October 2021*
- Explore sparse training in generative models.
- Study the surprising behavior of random pruning.
- Publication: Don't Be So Dense: Sparse-to-Sparse GAN Training Without Sacrificing Performance
- Publication: The Unreasonable Effectiveness of Random Pruning: Return of the Most Naive Baseline for Sparse Training

## PEER REVIEWED PUBLICATIONS

### Publication Summary

Overall: **40** papers (20 **A*** and 3 **A** conference paper, CORE Conference Ranking), **5** journal papers, and **13** preprints; among them **19** first-author papers and **4** last-author papers.
Impact: **406** citations, h-index: **11**, i10-index: **11** (as of June 12th 2023).

### Conference Publications (Sorted by date)

[**1**] Tianjin Huang*, **Shiwei Liu***, Tianlong Chen, Meng Fang, Li Shen, Vlado Menkovski, Lu Yin, Yulong Pei, Mykola Pechenizkiy. *Enhancing Adversarial Training via Reweighting Optimization Trajectory.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2023.

[**2**] Jiaxu Zhao, Lu Yin, **Shiwei Liu**, Meng Fang, Mykola Pechenizkiy. *REST: Debiasing Deep Neural Networks through Reweighted Sparse Training.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2023.

[**3**] Ajay Kumar Jaiswal, **Shiwei Liu**, Tianlong Chen, Ying Ding, Zhangyang Wang. *Instant Soup: Cheap Pruning Ensembles in A Single Pass Can Draw Lottery Tickets from Large Models.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023. **Oral**.

[**4**] Tianjin Huang, Lu Yin, Zhenyu Zhang, Li Shen, Meng Fang, Mykola Pechenizkiy, Zhangyang Wang, **Shiwei Liu**. *Are Large Kernels Better Teachers than Transformers for ConvNets?.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023.

[**5**] Ajay Kumar Jaiswal, **Shiwei Liu**, Tianlong Chen, Ying Ding, Zhangyang Wang. *Graph Ladling: Embarrassingly Scalable and Efficient Training of Powerful GNNs via Data-Centric Model Averaging.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023.

[**6**] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Xuxi Chen, Qiao Xiao, Boqian Wu, Mykola Pechenizkiy, Decebal Mocanu, and Zhangyang Wang. *More Convnets in the 2020s: Scaling Up Kernels*

*Beyond 51×51 Using Sparsity.* International Conference on Learning Representations (ICLR), 2023. Paper Code

[7] **Shiwei Liu***, Tianlong Chen*, Zhenyu Zhang, Xuxi Chen, Tianjin Huang, Ajay Kumar Jaiswal, and Zhangyang Wang. *Sparsity May Cry: Let Us Fail (Current) Sparse Neural Networks Together!.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-25% Spotlight**. Paper Code

[8] Duc N.M Hoang, **Shiwei Liu**, Radu Marculescu, Zhangyang Wang. *Revisiting Pruning at Initialization Through the Lens of Ramanujan Graph.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-5% Spotlight**. Paper Code

[9] Tianlong Chen, Zhenyu Zhang, Ajay Kumar Jaiswal, **Shiwei Liu**, Zhangyang Wang *Sparse MoE with Random Routing as the New Dropout: Training Bigger and Self-Scalable Models.* International Conference on Learning Representations (ICLR), 2023. **Notable-Top-25% Spotlight**.

[10] **Shiwei Liu**, and Zhangyang Wang., 2023. *Ten Lessons We Have Learned in the New "Sparseland": A Short Handbook for Sparse Neural Network Researchers.* SNN Workshop at ICLR 2023 **Spotlight**. Paper

[11] Lu Yin, Gen Li, Meng Fang, Li Shen, Tianjin Huang, Zhangyang Wang, Vlado Menkovski, Xiaolong Ma, Mykola Pechenizkiy, **Shiwei Liu**. Dynamic Sparsity Is Channel-Level Sparsity Learner. SNN Workshop at ICLR 2023 **Spotlight**.

[12] Lu Yin*, **Shiwei Liu***, Fang Meng, Tianjin Huang, Vlado Menkovski, Mykola Pechenizkiy. *Lottery Pools: Winning More by Interpolating Tickets without Increasing Training or Inference Cost.* Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), 2023.

[13] Tianjin Huang, Tianlong Chen, Meng Fang, Vlado Menkovski, Jiaxu Zhao, Lu Yin, Yulong Pei, Decebal Constantin Mocanu, Zhangyang Wang, Mykola Pechenizkiy, **Shiwei Liu**. *You Can Have Better Graph Neural Networks by Not Training Weights at All: Finding Untrained Graph Tickets.* Learning on Graphs Conference (LoG), 2022. **Best Paper Award**. Paper Code

[14] Qiao Xiao, Boqian Wu, Yu Zhang, **Shiwei Liu**, Mykola Pechenizkiy, Elena Mocanu, Decebal Constantin Mocanu. *Dynamic Sparse Network for Time Series Classification: Learning What to See.* The Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022.

[15] Lu Yin, Vlado Menkovski, Meng Fang, Tianjin Huang, Yulong Pei, Mykola Pechenizkiy, **Shiwei Liu**. *Superposing Many Tickets into One: A Performance Booster for Sparse Neural Network Training.* The 38th Conference on Uncertainty in Artificial Intelligence (UAI), 2022.

[16] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Li Shen, Decebal Constantin Mocanu, Zhangyang Wang, and Mykola Pechenizkiy. *The Unreasonable Effectiveness of Random Pruning: Return of the Most Naive Baseline for Sparse Training.* International Conference on Learning Representations (ICLR), 2022. Paper Code

[17] **Shiwei Liu**, Tianlong Chen, Zahra Atashgahi, Xiaohan Chen, Ghada Sokar, Elena Mocanu, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Deep Ensembling with No Overhead for either Training or Testing: The All-Round Blessings of Dynamic Sparsity.* International Conference on Learning Representations (ICLR), 2022. Paper Code

[18] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Zahra Atashgahi, Lu Yin, Huanyu Kou, Li Shen, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Sparse Training via Boosting Pruning Plasticity with Neuroregeneration.* The Thirty-fifth Conference on Neural Information Processing Systems (NeurIPS), 2021. Paper Code

[19] **Shiwei Liu**, Lu Yin, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *Do We Actually Need Dense Over-Parameterization? In-Time Over-Parameterization in Sparse Training.* The Thirty-eighth

International Conference on Machine Learning (ICML), PMLR, 2021. Paper Code

[20] **Shiwei Liu**, Decebal Constantin Mocanu, Yulong Pei, and Mykola Pechenizkiy. *Selfish sparse RNN training.* The Thirty-eighth International Conference on Machine Learning (ICML), PMLR, 2021.

[21] **Shiwei Liu**, Tim Van der Lee, Anil Yaman, Zahra Atashgahi, Davide Ferraro, Ghada Sokar, Mykola Pechenizkiy, and Decebal Constantin Mocanu. *Topological Insights into Sparse Neural Networks.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), Ghent, Belgium, 2020.

[22] **Shiwei Liu**. *Learning Sparse Neural Networks for Better Generalization.* $29^{th}$ International International Joint Conference on Artificial Intelligence (IJCAI), 2020. Doctoral Consortium.

[23] **Shiwei Liu**, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *On improving deep learning generalization with adaptive sparse connectivity.* The Workshop of Understanding and Improving Generalization in Deep Learning, The Thirty-sixth International Conference on Machine Learning (ICML), PMLR, 2019.

[24] Yan, Fulong, **Shiwei Liu**, and Nicola Calabretta. *Network Performance Optimization with Real Time Traffic Prediction in Data Center Network.* 2020 European Conference on Optical Communications (ECOC). IEEE, 2020.

**Journal Publications**

[25] **Shiwei Liu**, Yuesong Tian, Tianlong Chen, Li Shen. *Don't Be So Dense: Sparse-to-Sparse GAN Training Without Sacrificing Performance.* International Journal of Computer Vision (IJCV).

[26] Zahra Atashgahi, Xuhao Zhang, Neil Kichler, **Shiwei Liu**, Lu Yin, Mykola Pechenizkiy, Raymond Veldhuis, Decebal Constantin Mocanu. *Feature Selection with Neuron Evolution.*, TMLR. 2023.

[27] Zahra Atashgahi, Joost Pieterse, **Shiwei Liu**, Decebal Constantin Mocanu, Raymond Veldhuis, Mykola Pechenizkiy. *A Brain-inspired Algorithm for Training Highly Sparse Neural Networks.* Machine Learning Journal (ECML-PKDD 2022 journal track).

[28] **Shiwei Liu**, Decebal Constantin Mocanu, Amarsagar Reddy Ramapuram Matavalam, Yulong Pei, and Mykola Pechenizkiy. (2021). *Sparse evolutionary deep learning with over one million artificial neurons on commodity hardware.* Neural Computing and Applications, 1-16.

[29] **Shiwei Liu**, Iftitahu Nimah, Vlado Menkovski, Decebal Constantin Mocanu, and Mykola Pechenizkiy. (2021). *Efficient and effective training of sparse recurrent neural networks.* Neural Computing and Applications, 1-12.

**Pre-print and Under-Review**

[30] Boqian Wua, Qiao Xiaob, **Shiwei Liu**, Lu Yin, Mykola Pechenizkiy, Decebal Constantin Mocanu, Maurice Van Keulen, Elena Mocanu. *E2ENet: Dynamic Sparse Feature Fusion for Accurate and Efficient 3D Medical Image Segmentation.* Under Review.

[31] Jaiswal, Ajay, **Shiwei Liu**, Tianlong Chen, and Zhangyang Wang. *The Emergence of Essential Sparsity in Large Pre-trained Models: The Weights that Matter.* Under Review.

[32] **Shiwei Liu**, Tianjin Huang, Tianlong Chen, Zhangyang Wang. *The Counterattack of CNNs in Self-Supervised Learning: Larger Kernel Size Might be All You Need.* Under Review.

[33] Tiansheng Huang, **Shiwei Liu**, Li Shen, Fengxiang He, Weiwei Lin, Dacheng Tao., 2022. *Achieving Personalized Federated Learning with Sparse Local Models. arXiv preprint arXiv:2201.11380.* Paper

[34] Ruisi Cai, Xiaohan Chen, **Shiwei Liu**, Jayanth Srinivasa, Myungjin Lee, Ramana Rao Kompella, Zhangyang Wang. *Scaling Federated Learning under Data and Task Heterogeneity: A Pilot Study.*

Under Review.

[**35**] Hoang Pham, The-Anh Ta, **Shiwei Liu**, Lichuan Xiang, Dung D. Le, Hongkai Wen, Long Tran-Thanh. *Towards Data-Agnostic Pruning At Initialization: What Makes a Good Sparse Mask?* Under Review.

[**36**] Haoyu Ma, Chengming Zhang, Lizhi Xiang, Xiaolong Ma, Geng Yuan, Wenkai Zhang, **Shiwei Liu**, Tianlong Chen, Dingwen Tao, Yanzhi Wang, Zhangyang Wang, Xiaohui Xie. *HRBP: Hardware-friendly Regrouping towards Block-wise Pruning for Sparse Training.* Under Review.

[**37**] Yin Nan, Li Shen, Mengzhu Wang, **Shiwei Liu**, Chong Chen, Xian-Sheng Hua, Xiao Luo. *SPORT: A Subgraph Perspective on Graph Classification with Label Noise.* Under Review.

[**38**] Enneng Yang, Li Shen, Zhenyi Wang, **Shiwei Liu**, Guibing Guo, Xingwei Wang. *Data Augmented Flatness-aware Gradient Projection for Continual Learning.* Under Review.

[**39**] Zhangheng Li, **Shiwei Liu**, Tianlong Chen, Ajay Kumar Jaiswal, Zhenyu Zhang, Dilin Wang, Raghuraman Krishnamoorthi, Shiyu Chang, Zhangyang Wang. *Sparse Cocktail: Co-Training Many Sparsity Patterns and Ratios at Once.* Under Review.

**Ph.D. Thesis**

[**40**] **Shiwei Liu**. Sparse Neural Network Training with In-Time Over-Parameterization. The Eindhoven University of Technology. 2022. Paper

## SERVICE

2023: Area Chair: IJCAI, ICIP
Conference Reviewer: NeurIPS, ICML, ICLR, CVPR, ICCV, UAI, DAC
Journal Reviewer: JMLR, TPAMI

2022: Area Chair: ICIP
Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, AAAI.
Journal Reviewer: TPAMI

2021: Conference Reviewer: NeurIPS, ICLR, ICML, AISTATS.
Journal Reviewer: IEEE Transactions on Evolutionary Computation.

2020: Conference Reviewer: ECMLPKDD, IDA.
Journal Reviewer: ACM Transactions On Intelligent Systems And Technology.

## EDUCATIONAL AND TEACHING ACTIVITIES

**Tutorial Representation**

• IJCAI 2023, "Sparse Training for Supervised, Unsupervised, Continual, and Deep Reinforcement Learning with Deep Neural Networks", Macao, China. Link Aug. 2023
Co-organizer & Presenter; one-hour tutorial presentation from sparse training to sparse scaling.

• ECMLPKDD 2022, "Sparse Neural Network Training", Grenoble, France. Link Sep. 2022
Co-organizer & Presenter; 45-min tutorial presentation on understanding sparse neural network training with supervised learning.

**Students Mentoring**

PhD Student - Research Advisor:

• Lu Yin, Eindhoven University of Technology Dec. 2021 - Current

• Tianjin Huang, Eindhoven University of Technology Dec. 2021 - Current

• Boqian Wu, University of Twente Aug. 2022 - Current

- Qiao Xiao, Eindhoven University of Technology — Aug. 2021 - Current
- Ajay Kumar Jaiswal, University of Texas at Austin — Sep. 2022 - Current
- Duc N.M Hoang, University of Texas at Austin — Sep. 2022 - Current
- Zhenyu Zhang, University of Texas at Austin — Sep. 2022 - Current

Master Student - Research Advisor:

- Tiansheng Huang, South China University of Technology — Jun. 2021 - Dec. 2021

## MEDIA COVERAGE

You Can Have Better Graph Neural Networks by Not Training Weights at All: Finding Untrained GNNs Tickets. UT Austin News — Jan. 2023

## HOBBY

Fitness, Rock Music, Travelling